Alcohols, carboxylic acids, esters and polymers

30 questions: 10 multiple choice, 20 structured.

---

1 · Part 2 **Paper 2 only** Alcohols [1]

Which formula represents propan-1-ol?

A $CH_3CH_2CH_3$

B $CH_3CH_2CH_2OH$

C CH_3CH_2COOH

D $CH_3CH_2CH_2CH_2OH$

---

2 · Part 2 **Paper 2 only** Alcohols [1]

Which set of conditions is used in industry to manufacture ethanol from ethene and steam?

	catalyst	temperature	pressure
A	yeast	30 °C	1 atm
B	phosphoric acid	30 °C	60–70 atm
C	silica or alumina	650 °C	60–70 atm
D	phosphoric acid	300 °C	60–70 atm

Topical Hub
topical-hub.pages.dev

3 · Part 2 Paper 2 only Alcohols [4]

Alcohols are a homologous series of compounds that contain the –OH functional group.

(a) The table shows information about two alcohols. Complete the table.

name	structural formula	molecular formula
methanol		CH_4O
	$CH_3CH_2CH_2CH_2OH$	

[3]

(b) Deduce the molecular formula of the unbranched alcohol that has five carbon atoms in each molecule. [1]

Topical Hub
topical-hub.pages.dev

4 · Part 2

Paper 2 only

Alcohols

[6]

Ethanol can be made by the fermentation of glucose solution using yeast. The fermentation is carried out in the absence of air at a temperature of about 30 °C.

(a) Yeast contains biological catalysts that speed up fermentation. Give the name of these biological catalysts. [1]

(b) Explain why fermentation is carried out at about 30 °C, rather than at 10 °C or at 60 °C. [3]

(c) Explain why air must be kept out of the fermentation vessel. [2]

Topical Hub
topical-hub.pages.dev

5 · Part 2

Paper 2 only

Alcohols

[7]

The equation for the fermentation of glucose is shown.

$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$

Use the Periodic Table to help you.

(a) Calculate the maximum mass of ethanol that can be made from 45.0 g of glucose. [3]

(b) In an experiment, 45.0 g of glucose produced 18.4 g of ethanol. Calculate the percentage yield of ethanol. [2]

(c) Calculate the volume of carbon dioxide, in dm^3 at rtp, produced when 45.0 g of glucose is completely fermented. (1 mol of any gas occupies 24 dm^3 at rtp) [2]

Topical Hub
topical-hub.pages.dev

6 · Part 2

Paper 2 only

Alcohols

[7]

Ethanol is used as a fuel. Burning ethanol in a plentiful supply of air is an oxidation reaction.

(a) Complete the balanced equation for the complete combustion of ethanol.

C_2H_5OH + ... → ..... + ...

[2]

(b) Ali uses a spirit burner containing ethanol to heat some water in a copper can. The table shows Ali's results. Calculate the enthalpy change, in kJ/mol, for the combustion of ethanol. The specific heat capacity of water is 4.2 J/g/°C.

measurement	**value**
mass of spirit burner and ethanol at the start	186.32 g
mass of spirit burner and ethanol at the end	185.40 g
mass of water in the copper can	100 g
starting temperature of the water	19.5 °C
highest temperature of the water	49.5 °C

[4]

(c) The accepted value for the enthalpy change of combustion of ethanol is −1367 kJ/mol. Give one reason why Ali's value is less negative than the accepted value.

[1]

Topical Hub
topical-hub.pages.dev

7 · Part 2

Paper 2 only

Alcohols

[5]

Ethanol can be oxidised to ethanoic acid.

(a) State the reagent and the condition used to oxidise ethanol to ethanoic acid in the laboratory. [2]

(b) Wine contains ethanol. Wine that is left in an open bottle for several days starts to taste sour. Explain this observation. [2]

(c) Suggest why keeping an opened bottle of wine tightly sealed slows down this change. [1]

Topical Hub
topical-hub.pages.dev

8 · Part 2

Paper 2 only | Carboxylic acids | [1]

What is the name of the compound with the structural formula $CH_3CH_2CH_2COOH$?

A butanoic acid

B propanoic acid

C butan-1-ol

D ethyl ethanoate

9 · Part 2

Paper 2 only | Carboxylic acids | [1]

Which carboxylic acid is present in vinegar?

A methanoic acid

B butanoic acid

C ethanoic acid

D propanoic acid

Topical Hub
topical-hub.pages.dev

10 · Part 2

Paper 2 only Carboxylic acids [5]

Carboxylic acids form a homologous series.

(a) The table shows information about two carboxylic acids. Complete the table.

name	structural formula
	HCOOH
propanoic acid	

[2]

(b) Explain why the two acids in the table are members of the same homologous series. [2]

(c) Predict the name of the unbranched carboxylic acid that has five carbon atoms in each molecule. [1]

Topical Hub
topical-hub.pages.dev

11 · Part 2

Paper 2 only

Carboxylic acids

[7]

Limescale, which is mainly calcium carbonate, builds up inside kettles. It can be removed by adding a solution of ethanoic acid.

(a) Give two observations you would expect to make when ethanoic acid solution is added to calcium carbonate. [2]

(b) Complete the balanced equation for the reaction. State symbols are not required.

$CaCO_3$ + ...CH_3COOH → $(CH_3COO)_2Ca$ + ... + ... [2]

(c) In one kettle, 0.0400 mol of ethanoic acid reacts completely with calcium carbonate. Calculate the mass of calcium carbonate that reacts. Use the Periodic Table to help you. [3]

Topical Hub
topical-hub.pages.dev

12 · Part 2

Paper 2 only

Carboxylic acids

[6]

Zinc reacts with an aqueous solution of ethanoic acid.

$2CH_3COOH(aq) + Zn(s) \rightarrow (CH_3COO)_2Zn(aq) + H_2(g)$

(a) Give the name of the salt with the formula $(CH_3COO)_2Zn$. [1]

(b) Kemi adds 0.975 g of zinc to an excess of ethanoic acid solution. Calculate the volume of hydrogen, in cm^3 at rtp, produced when all the zinc reacts. (1 mol of any gas occupies 24 dm^3 at rtp) [3]

(c) Describe the test for hydrogen gas and give the result. [2]

Topical Hub
topical-hub.pages.dev

13 · Part 2 Paper 2 only Esters [1]

Which compound is an ester?

A $CH_3CH_2CH_2CH_2COOH$

B $CH_3CH_2COOCH_3$

C $CH_3CH_2CH_2CH_2OH$

D $CH_3CH=CHCH_3$

14 · Part 2 Paper 2 only Esters [1]

Which row gives the correct name and structural formula of the ester made from methanol and butanoic acid?

	name of ester	structural formula of ester
A	methyl butanoate	$CH_3CH_2CH_2COOCH_3$
B	butyl methanoate	$HCOOCH_2CH_2CH_2CH_3$
C	methyl butanoate	$HCOOCH_2CH_2CH_2CH_3$
D	butyl methanoate	$CH_3CH_2CH_2COOCH_3$

TopicalHub topicalhub.pages.dev

15 · Part 2 Paper 2 only Esters [1]

Which functional group is present in all esters?

A –OH

B –COOH

C >C=C<

D –COO–

16 · Part 2

Paper 2 only

Esters

[5]

Ethyl ethanoate is an ester. It is used as a solvent in some nail-varnish removers.

(a) Give the names of the alcohol and the carboxylic acid that react together to form ethyl ethanoate. [2]

(b) Give the structural formula of ethyl ethanoate. [1]

(c) Give the name of the other product of this reaction. [1]

(d) Give the molecular formula of ethyl ethanoate. [1]

Topical Hub
topical-hub.pages.dev

17 · Part 2

Paper 2 only

Esters

[6]

Mia makes a sample of the ester propyl ethanoate. This is her method.

- Mix 5 cm^3 of propan-1-ol with 5 cm^3 of ethanoic acid in a boiling tube.
- Add five drops of concentrated sulfuric acid.
- Warm the boiling tube in a beaker of hot water for ten minutes.
- Pour the mixture into a beaker of sodium carbonate solution.

(a) State the role of the concentrated sulfuric acid. [1]

(b) Explain why Mia warms the boiling tube in a beaker of hot water instead of heating it directly with a Bunsen burner flame. [2]

(c) Suggest why Mia pours the mixture into sodium carbonate solution. [2]

(d) State how Mia could tell that an ester has formed. [1]

Topical Hub
topicalhub.pages.dev

18 · Part 2

Paper 2 only

Esters

[7]

Esters are made by reacting an alcohol with a carboxylic acid.

(a) The table shows information about two esters. Complete the table.

alcohol	carboxylic acid	name of ester	structural formula of ester
methanol	ethanoic acid		
		ethyl propanoate	

[5]

(b) An ester has the structural formula $CH_3COOCH_2CH_2CH_2CH_3$. Give the names of the alcohol and the carboxylic acid used to make this ester. [2]

Topical Hub
topical-hub.pages.dev

19 · Part 2

Paper 2 only

Esters

[7]

Ethyl butanoate is an ester. It is made by reacting ethanol with butanoic acid in the presence of an acid catalyst.

Topical Hub
topical-hub.pages.dev

(a) Give the structural formula of ethyl butanoate. [1]

(b) The molecular formula of ethyl butanoate is $C_6H_{12}O_2$. Calculate its relative formula mass (Mr). Use the Periodic Table to help you. [1]

(c) Mai reacts 8.80 g of butanoic acid (Mr = 88) with an excess of ethanol. One mole of butanoic acid forms one mole of ethyl butanoate. Calculate the maximum mass of ethyl butanoate that Mai could make. [2]

(d) Mai obtains 8.12 g of ethyl butanoate. Calculate the percentage yield. [2]

(e) Suggest one reason why the percentage yield is less than 100%. [1]

Topical Hub
topical-hub.pages.dev

20 · Part 2

Paper 2 only

Esters

[5]

Some esters are used in perfumes.

(a) Give two properties of esters that make them suitable for use in perfumes. [2]

(b) The table shows the boiling points of three esters. Deduce which ester in the table evaporates most easily at room temperature. Give a reason for your answer.

ester	boiling point (°C)
ethyl methanoate	54
ethyl ethanoate	77
methyl butanoate	102

[2]

(c) State one use of esters in the food industry. [1]

Topical Hub
topical-hub.pages.dev

21 · Part 2 Paper 1 & 2 Synthetic polymers [1]

Which compound can be used as a monomer to make an addition polymer?

A $CH_3CH_2CH_3$

B CH_3CH_2OH

C $CH_2=CHCH_2CH_3$

D CH_3COOH

22 · Part 2 Paper 1 & 2 Synthetic polymers [1]

The repeat unit of addition polymer W is $-[CH_2-CCl_2]-n$.

What is the structural formula of the monomer used to make W?

A $CH_2=CCl_2$

B CH_3CCl_3

C CHCl=CHCl

D $CH_2=CHCl$

23 · Part 2 Paper 2 only Synthetic polymers [1]

What is formed when many molecules of a dicarboxylic acid react with many molecules of a diol?

A a polyester only

B a polyester and water

C an addition polymer and water

D an addition polymer only

Topical Hub
topical-hub.pages.dev

24 · Part 2

Paper 1 & 2 Synthetic polymers [5]

Addition polymers are made from monomers that contain a carbon–carbon double bond.

(a) The table shows information about three monomers and the addition polymers made from them. Complete the table. Write each repeat unit in the same style as the one given.

name of monomer	formula of monomer	repeat unit of polymer
ethene	$CH_2=CH_2$	
chloroethene		$-[CH_2-CHCl]-n$
tetrafluoroethene	$CF_2=CF_2$	

[3]

(b) Explain why ethane, CH_3CH_3, cannot be used to make an addition polymer. [2]

Topical Hub
topical-hub.pages.dev

25 · Part 2

Paper 1 & 2 | Synthetic polymers | [5]

Poly(chloroethene) is an addition polymer used to make water pipes and window frames. When these items are no longer needed, they are often buried in landfill sites or burned.

(a) Explain the problems caused by disposing of waste poly(chloroethene) in landfill sites and by burning it. [4]

(b) Suggest one other way of dealing with waste poly(chloroethene) items that reduces these problems. [1]

Topical Hub
topical-hub.pages.dev

26 · Part 2

Paper 1 & 2 | Synthetic polymers | [6]

Poly(propene) is an addition polymer made from propene, $CH_2=CHCH_3$.

(a) Give the repeat unit of poly(propene). [1]

(b) Calculate the relative formula mass (Mr) of propene. Use the Periodic Table to help you. [1]

(c) A sample of poly(propene) has an average relative formula mass of 126 000. Calculate the average number of propene molecules that join together to form one polymer molecule. [2]

(d) Explain why the relative formula mass of the repeat unit of poly(propene) is the same as the relative formula mass of propene. [2]

Topical Hub
topical-hub.pages.dev

27 · Part 2

Paper 1 & 2 | Synthetic polymers | [5]

This question is about three addition polymers, X, Y and Z.

(a) The repeat unit of polymer X is –[CF_2–CFCl]–n. Give the structural formula of the monomer used to make X. [1]

(b) Polymer Y is made from the monomer but-2-ene, $CH_3CH=CHCH_3$. Give the repeat unit of polymer Y. [1]

(c) Part of a molecule of polymer Z is shown. Deduce the repeat unit of polymer Z and the structural formula of its monomer.

–CH_2–CHF–CH_2–CHF–CH_2–CHF– [2]

(d) State what is meant by the term monomer. [1]

Topical Hub
topical-hub.pages.dev

28 · Part 2

Paper 2 only

Synthetic polymers

[5]

Ethanedioic acid, HOOC–COOH, reacts with ethanediol, $HOCH_2CH_2OH$, to form a polyester.

(a) State the type of polymerisation that takes place. [1]

(b) Give the repeat unit of the polyester formed. [2]

(c) Explain why each monomer molecule must have two functional groups for a polymer to form. [2]

Topical Hub
topical-hub.pages.dev

29 · Part 2

Paper 2 only

Synthetic polymers

[5]

The table shows the monomers used to make two polymers, R and S.

polymer	monomer or monomers
R	$CH_2=C(CH_3)_2$
S	$HOOC–CH_2CH_2–COOH$ and $HOCH_2CH_2CH_2OH$

(a) Polymer R is an addition polymer. Give the repeat unit of polymer R. [1]

(b) Polymer S is a polyester. Give the repeat unit of polymer S. [2]

(c) Polymer S is a biopolyester. Explain why polymer S causes fewer problems than polymer R when they are buried in landfill sites. [2]

Topical Hub
topical-hub.pages.dev

30 · Part 2

Paper 2 only

Synthetic polymers

[6]

Poly(ethene) and polyesters are both synthetic polymers, but they are formed by different types of polymerisation.

(a) Explain the differences between the polymerisation that forms poly(ethene) from ethene and the polymerisation that forms a polyester from its monomers. [5]

(b) Give one way in which the two types of polymerisation are similar. [1]

Topical Hub
topical-hub.pages.dev